

GAIRA V6 — Theme Optimisation

Why 13 chemical themes, and how that number was chosen

The number of themes is a **design choice with a measurable cost**. Fewer themes are easier to recover and harder to interpret; more themes are the reverse. This report sweeps 16 levels $\times$ 5 generation methods = 70 candidate hierarchies, scores every one against a permutation null, and selects on the Pareto front of recoverability $\times$ interpretability — broken, where the score is flat, by a pre-stated chemical constraint.

1 · The two axes

Recoverability must be chance-corrected, because raw top-1 accuracy rises mechanically as K falls (at $K = 2$ a coin is right half the time):

$$\kappa = (\text{top1} - \text{null_top1}) / (1 - \text{null_top1})$$

The null permutes the analyte $\rightarrow$ expected-theme assignment 2000 times and recomputes the same statistic, so it absorbs both the number of themes and the very uneven prior mass of each theme.

Interpretability combines chemical coherence, spectral coherence and resolution:

$$I = 0.4 \cdot C_{\text{chem}} + 0.3 \cdot C_{\text{spec}} + 0.3 \cdot (\log K / \log K_{\text{max}})$$

C_{chem} is 1 — the normalised entropy of the chemical families of a theme's member analytes; C_{spec} is the mean pairwise cosine between member motifs' implied Raman spectra. The resolution term is essential: without it a 2-theme hierarchy would score as maximally interpretable for saying almost nothing.

Part 4 — theme optimisation: accuracy alone is the wrong objective

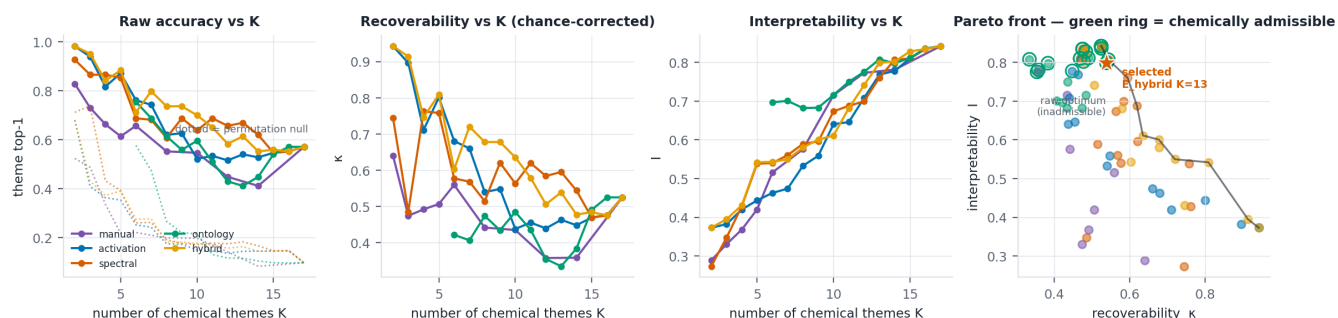


Figure 1 — Left to right: raw accuracy against K (dotted = permutation null, and note how it tracks the accuracy); chance-corrected recoverability; interpretability; and the Pareto front with chemically admissible partitions ringed in green.

2 · The Pareto front

method	K	top-1	null	κ	I	$\kappa \times I$	admissible
C_spectral	13	0.669	0.182	0.595	0.760	0.4523	NO
B_activation	17	0.571	0.096	0.525	0.842	0.4420	yes
C_spectral	14	0.620	0.166	0.544	0.807	0.4389	yes
E_hybrid	5	0.883	0.390	0.809	0.542	0.4385	NO
C_spectral	11	0.687	0.178	0.619	0.688	0.4262	NO
E_hybrid	9	0.736	0.181	0.678	0.601	0.4071	NO
E_hybrid	7	0.797	0.277	0.720	0.550	0.3959	NO
E_hybrid	10	0.699	0.176	0.635	0.611	0.3883	NO
E_hybrid	3	0.951	0.431	0.914	0.395	0.3609	NO
B_activation	2	0.982	0.680	0.943	0.373	0.3514	NO
E_hybrid	2	0.982	0.680	0.943	0.373	0.3514	NO

3 · Breaking a flat tie

The composite spans only 0.43–0.45 across four candidates at K = 13, 14 and 17 from three different methods. At n = 163 scored analytes that band is statistically indistinguishable, so the score does not select. A third decimal place is not a scientific argument.

Pre-stated constraint. A theme must span at most one non-bridging chemical superclass. Superclasses fixed on chemistry before scoring; phosphate_ester is the only BRIDGING class.

Superclass map (fixed on chemistry before any partition was scored):

motif chemical class	superclass
phosphate_ester	BRIDGING
monosaccharide	carbohydrate
polysaccharide	carbohydrate
polyene	cofactor
redox_cofactor	cofactor
tetrapyrrole	cofactor
acylglycerol	lipid
fatty_acid	lipid
sterol	lipid
organic_acid	metabolite
sulfur_metabolite	metabolite
nucleobase_purine	nucleic
nucleobase_pyrimidine	nucleic
aromatic_sidechain	protein
free_amino_acid	protein
polypeptide	protein

Only **21 of 70** partitions survive. The raw-score optimum — **C_spectral at K=13** — does not: it places *polysaccharide_glycosidic* in the same theme as *protein_amide_backbone*. That is a real spectral overlap (both are broad in the CH₂ / amide-III region) but it is not a biochemical class, and a theme that cannot be named is not a theme.

Within the admissible band the smallest K is preferred, because a theme layer that does not abstract is only the motif layer relabelled.

4 • The selected hierarchy

chemical theme	member MSS motifs
Purine	oxopurine_carbonyl, purine_ring_breathing
Pyrimidine	pyrimidine_ring
Monosaccharide + Phosphate	glycan_co_network, nucleic_backbone_phosphate
Aromatic residue	aromatic_ring_residue
Protein backbone	protein_amide_backbone
Free amino acid	amino_acid_zwitterion
Acyl lipid	fatty_acyl_chain, triglyceride_ester
Sterol	sterol_ring_system
Polysaccharide	polysaccharide_glycosidic
Organic acid	carboxylate_organic_acid
Sulfur metabolite	sulfur_heterocycle_thione
Flavin + Porphyrin	flavin_redox_cofactor, porphyrin_macrocycle
Carotenoid	carotenoid_polyene

E_hybrid at K = 13. Four merges, each chemically defensible: purine ring + oxopurine carbonyl (both purine scaffolds); monosaccharide + phosphate ester (sugar phosphates); fatty acyl + acylglycerol (both acyl chains, distinguished only by the ester); flavin + porphyrin (both conjugated N-heterocyclic cofactors, both corpus-limited). Sterol, polysaccharide, pyrimidine, aromatic residue, protein backbone, free amino acid, organic acid, sulfur and carotenoid remain separate.

5 • Sensitivity

The table below is the full sweep. Two patterns are worth stating plainly. First, **κ falls as K rises** for every method — finer hierarchies are genuinely harder, and no amount of clustering changes that. Second, **the null tracks the accuracy closely** at low K, which is exactly why the raw curve must not be read as performance.

method	K	top-1	null	κ	macro F1	ECE	I	$\kappa \times I$	adm
A_manual	2	0.828	0.523	0.640	0.815	0.179	0.289	0.1848	
A_manual	3	0.730	0.487	0.474	0.649	0.168	0.330	0.1566	
A_manual	4	0.663	0.337	0.491	0.557	0.224	0.368	0.1809	
A_manual	5	0.614	0.217	0.506	0.586	0.191	0.420	0.2128	
A_manual	6	0.656	0.221	0.559	0.536	0.247	0.516	0.2884	
A_manual	8	0.552	0.199	0.441	0.361	0.172	0.576	0.2540	
A_manual	10	0.546	0.197	0.434	0.341	0.169	0.715	0.3108	
A_manual	12	0.448	0.141	0.357	0.293	0.112	0.773	0.2760	
A_manual	14	0.411	0.083	0.358	0.347	0.159	0.782	0.2798	y
A_manual	17	0.571	0.097	0.524	0.438	0.347	0.842	0.4415	y
B_activation	2	0.982	0.680	0.943	0.953	0.147	0.373	0.3514	
B_activation	3	0.939	0.409	0.896	0.876	0.336	0.382	0.3427	
B_activation	4	0.816	0.363	0.711	0.706	0.263	0.420	0.2987	

method	K	top-1	null	κ	macro F1	ECE	I	$\kappa \times I$	adm
B_activation	5	0.871	0.353	0.801	0.680	0.370	0.444	0.3552	
B_activation	6	0.761	0.252	0.680	0.630	0.316	0.463	0.3149	
B_activation	7	0.742	0.242	0.660	0.600	0.326	0.474	0.3128	
B_activation	8	0.620	0.175	0.539	0.515	0.247	0.533	0.2875	
B_activation	9	0.626	0.172	0.548	0.487	0.258	0.559	0.3060	
B_activation	10	0.521	0.149	0.437	0.429	0.175	0.641	0.2802	
B_activation	11	0.534	0.145	0.455	0.406	0.186	0.646	0.2939	
B_activation	12	0.515	0.136	0.439	0.384	0.177	0.708	0.3110	
B_activation	13	0.540	0.144	0.463	0.421	0.208	0.769	0.3555	
B_activation	14	0.528	0.145	0.448	0.391	0.207	0.778	0.3484	y
B_activation	15	0.546	0.145	0.469	0.389	0.234	0.810	0.3799	y
B_activation	16	0.552	0.145	0.476	0.427	0.247	0.835	0.3973	y
B_activation	17	0.571	0.096	0.525	0.438	0.347	0.842	0.4420	y
C_spectral	2	0.926	0.712	0.745	0.801	0.232	0.273	0.2035	
C_spectral	3	0.865	0.738	0.485	0.531	0.232	0.347	0.1684	
C_spectral	4	0.865	0.432	0.762	0.632	0.324	0.428	0.3266	
C_spectral	5	0.853	0.392	0.758	0.630	0.339	0.538	0.4081	
C_spectral	6	0.687	0.260	0.577	0.578	0.237	0.540	0.3117	
C_spectral	7	0.681	0.262	0.568	0.524	0.234	0.561	0.3184	
C_spectral	8	0.607	0.192	0.514	0.543	0.201	0.590	0.3033	
C_spectral	9	0.687	0.176	0.620	0.582	0.319	0.596	0.3696	
C_spectral	10	0.638	0.172	0.563	0.476	0.277	0.674	0.3794	
C_spectral	11	0.687	0.178	0.619	0.514	0.333	0.688	0.4262	
C_spectral	12	0.656	0.175	0.584	0.411	0.308	0.700	0.4084	
C_spectral	13	0.669	0.182	0.595	0.409	0.326	0.760	0.4523	
C_spectral	14	0.620	0.166	0.544	0.466	0.293	0.807	0.4389	y
C_spectral	15	0.546	0.146	0.469	0.389	0.234	0.810	0.3797	y
C_spectral	16	0.552	0.147	0.475	0.427	0.247	0.835	0.3966	y
C_spectral	17	0.571	0.097	0.524	0.438	0.347	0.842	0.4416	y
D_ontology	6	0.755	0.576	0.421	0.254	0.201	0.697	0.2935	
D_ontology	7	0.687	0.473	0.406	0.271	0.196	0.701	0.2845	
D_ontology	8	0.614	0.266	0.473	0.328	0.269	0.682	0.3229	
D_ontology	9	0.558	0.220	0.434	0.291	0.231	0.683	0.2964	
D_ontology	10	0.595	0.215	0.484	0.295	0.285	0.716	0.3469	
D_ontology	11	0.509	0.131	0.435	0.378	0.221	0.750	0.3265	
D_ontology	12	0.429	0.116	0.354	0.397	0.147	0.776	0.2749	y
D_ontology	13	0.411	0.115	0.334	0.395	0.138	0.808	0.2701	y
D_ontology	14	0.448	0.104	0.383	0.465	0.203	0.798	0.3061	y
D_ontology	15	0.540	0.097	0.490	0.465	0.301	0.811	0.3981	y
D_ontology	16	0.571	0.097	0.525	0.465	0.341	0.836	0.4388	y
D_ontology	17	0.571	0.097	0.525	0.438	0.347	0.842	0.4418	y
E_hybrid	2	0.982	0.680	0.943	0.953	0.147	0.373	0.3514	

method	K	top-1	null	κ	macro F1	ECE	I	$\kappa \times I$	adm
E_hybrid	3	0.951	0.431	0.914	0.892	0.344	0.395	0.3609	
E_hybrid	4	0.841	0.372	0.746	0.720	0.291	0.431	0.3215	
E_hybrid	5	0.883	0.390	0.809	0.692	0.352	0.542	0.4385	
E_hybrid	6	0.712	0.275	0.602	0.626	0.239	0.543	0.3272	
E_hybrid	7	0.797	0.277	0.720	0.657	0.367	0.550	0.3959	
E_hybrid	8	0.736	0.182	0.677	0.679	0.350	0.580	0.3931	
E_hybrid	9	0.736	0.181	0.678	0.621	0.358	0.601	0.4071	
E_hybrid	10	0.699	0.176	0.635	0.533	0.337	0.611	0.3883	
E_hybrid	11	0.650	0.170	0.579	0.445	0.297	0.682	0.3946	
E_hybrid	12	0.583	0.156	0.505	0.434	0.245	0.741	0.3748	
E_hybrid	13	0.614	0.163	0.538	0.501	0.282	0.799	0.4301	y
E_hybrid	14	0.552	0.143	0.477	0.420	0.234	0.802	0.3831	y
E_hybrid	15	0.558	0.145	0.484	0.458	0.248	0.829	0.4007	y
E_hybrid	16	0.552	0.147	0.475	0.427	0.247	0.835	0.3965	y
E_hybrid	17	0.571	0.097	0.524	0.438	0.347	0.842	0.4415	y

Atlas 09ed804a40836f4a05a91ba10900cded unchanged. Reproduce: python results/v6_rebuild/code/p03_theme_optimisation.py